

Combinatorial Optimization Using Ant Colony Optimization

Homework 7: 15-Department Quadratic Assignment Problem

1. Introduction

In this project, I solved the Quadratic Assignment Problem (QAP) using an Ant Colony Optimization (ACO) algorithm. ACO, introduced by Marco Dorigo in the early 1990s, is a population-based constructive metaheuristic inspired by the foraging behavior of real ants [1]. Unlike the single-solution methods such as tabu search studied previously in this course, ACO builds solutions incrementally by having artificial ants construct complete solutions one component at a time, guided by pheromone trails and a problem-specific heuristic [2].

The core mechanism of ACO relies on two information sources: pheromone values that encode the collective experience of the colony, and a local heuristic that provides problem-specific guidance. At each construction step, an ant selects the next component probabilistically by weighing these two factors. After all ants complete their solutions, pheromone trails are updated: first evaporated globally, then reinforced along the paths of successful ants. This positive feedback loop encourages future ants to explore promising regions of the search space while the stochastic selection mechanism maintains diversity [3].

The probability of ant selecting arc (i, j) is given by: $p_{i,j} = (\tau_{i,j}^{\alpha} * \eta_{i,j}^{\beta}) / \text{Sum}(\tau_{i,k}^{\alpha} * \eta_{i,k}^{\beta})$, where $\tau_{i,j}$ is the pheromone intensity, $\eta_{i,j}$ is the local heuristic value, α controls the pheromone importance, and β controls the heuristic importance.

For this homework, the problem to solve is the 15-department QAP (Nugent Nug15 benchmark), where 15 departments must be placed in a 3 by 5 grid to minimize total flow cost. The known optimal solution has a cost of 1150 [4]. In this report, the ACO algorithm setup is described, followed by systematic experimentation with different parameter configurations, and finally the results are analyzed with conclusions about the algorithm behavior on this problem.

2. Methodology, Definitions and Assumptions

2.1. Problem definition

The optimization target is the 15-department Quadratic Assignment Problem from Nugent et al. (1968), known as Nug15 [4]. Fifteen departments must be assigned to 15 locations arranged in a 3 x 5 grid. The objective is to minimize the total flow cost, calculated as the sum of flow times distance for all pairs of departments. Both flow and distance matrices are symmetric. The known optimal solution has a cost of 1150 (using the symmetric flow x distance formulation).

2.2. Solution encoding

Each solution is encoded as a permutation of integers [0, 1, ..., 14], where the index represents the department and the value at that index represents the assigned location. This encoding was chosen because it provides a natural one-to-one mapping between departments and locations, ensures feasibility (every department gets exactly one location), and allows the constructive nature of ACO to build valid solutions incrementally.

2.3. Solution construction

Each ant constructs a complete solution by assigning departments to locations one at a time. At each construction step, the ant considers all unassigned locations for the current department and selects one probabilistically based on the pheromone intensity and local heuristic value. The order in which departments are processed is randomized for each ant to introduce additional diversity. This randomized construction order is an important design choice that prevents all ants from making the same initial assignments and helps explore different regions of the search space.

2.4. Local heuristic

The local heuristic $\eta_{i,j}$ for assigning department i to location j is defined as: $\eta_{i,j} = 1 / (1 + \text{partial_cost})$, where partial_cost is the sum of $\text{flow}[i][k] * \text{dist}[j][\text{loc}_k]$ for all departments k that have already been assigned to location loc_k . This heuristic measures the incremental cost of placing department i at location j given the current partial solution. A lower partial cost results in a higher heuristic value, guiding ants toward locally promising placements. This heuristic directly reflects the objective function, making it a natural choice for the QAP.

2.5. Pheromone structure

The pheromone matrix τ is a 15 x 15 matrix where $\tau[i][j]$ represents the desirability of assigning department i to location j . The pheromone is initialized uniformly to a value τ_0 . After each iteration, pheromone is updated in two phases: (1) global evaporation where all pheromone values are multiplied by $(1 - \rho)$, and (2) pheromone deposit where ants reinforce arcs in their solutions. The deposit amount for each ant is $1000 / \text{cost}$, which scales inversely with solution quality to reward better solutions with more pheromone.

2.6. Stopping criterion

All runs use a fixed stopping criterion of 200 iterations. This value was selected to provide sufficient time for the constructive nature of ACO to converge while keeping computational costs manageable across all 60 experimental runs.

2.7. Experimental design

To systematically evaluate the ACO design choices, nine experiments are conducted with five random seeds each (42, 123, 256, 789, 1024) for a total of 60 runs. The experiments progressively build on each other, with the best-performing parameter from each experiment carried forward to subsequent ones. The experiments are summarized in the following table.

Table 1: Detail of the experiments (60 total runs across 9 ACO variants)

Experiment	Ant s	α	β	ρ	τ_0	Special	Runs
E1: Base ACO	10	1	2	0. 5	1.0	None	5
E2: Small pop	5	1	2	0. 5	1.0	None	5
E2: Large pop	20	1	2	0. 5	1.0	None	5
E3: Low pheromone	20	1	2	0. 5	0.1	None	5
E3: High pheromone	20	1	2	0. 5	5.0	None	5
E4: Low evaporation	20	1	2	0. 2	5.0	None	5
E4: High evaporation	20	1	2	0. 8	5.0	None	5
E5: Heuristic only	20	0	2	0. 8	5.0	alpha=0	5
E6: Pheromone only	20	1	0	0. 8	5.0	beta=0	5
E7: q_0 exploit.	20	1	2	0. 8	5.0	$q_0=0.5$	5
E8: Local update	20	1	2	0. 8	5.0	Local pheromone	5
E9: Elitist	20	1	2	0. 8	5.0	Elitist 30% + local	5

3. Results and Discussion

3.1. Preliminary runs (Experiment 1: Base ACO)

3.1.1. Parameter definition

The preliminary runs establish a baseline for the ACO algorithm on the Nug15 QAP. The base configuration uses 10 ants, $\alpha = 1$, $\beta = 2$, evaporation rate $\rho = 0.5$, and initial pheromone $\tau_0 = 1.0$. The β value of 2 was chosen to give the local heuristic slightly more influence than the pheromone, which is common practice for constructive heuristics where the problem-specific knowledge is valuable in the early stages of the search. Five seeds are used to test sensitivity to randomness.

3.1.2. Results

The following figures show the near-optimums found by the base ACO for each seed and the convergence behavior over 200 iterations.

Table 2: Experiment 1 results - Base ACO with 5 seeds

Seed	Best Cost	Gap from Optimal	Runtime (s)
42	1292	142	0.70
123	1282	132	0.71
256	1282	132	0.72
789	1260	110	0.70
1024	1290	140	0.70

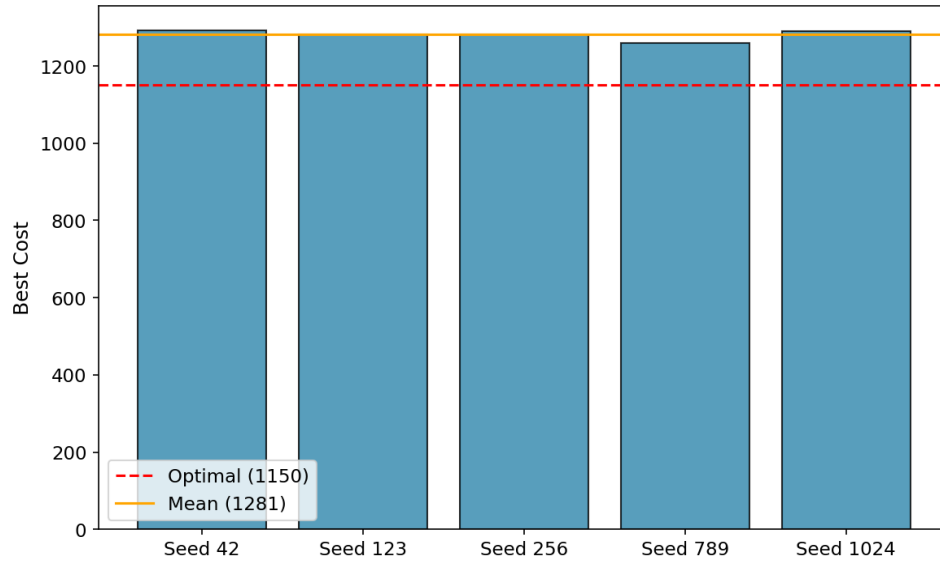


Figure 1: Near-optimums reported by seed for the preliminary runs

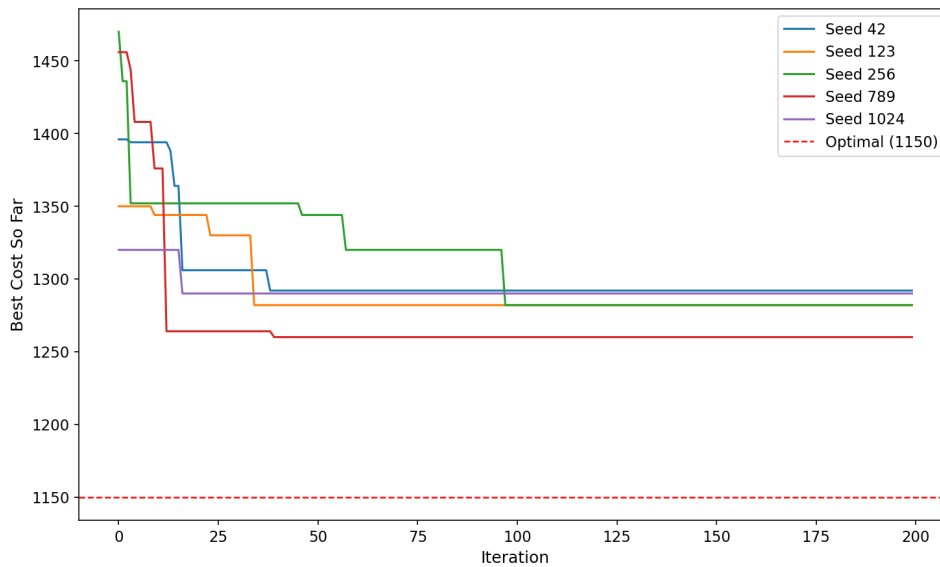


Figure 2: Convergence of best cost over iterations (Experiment 1)

3.1.3. Conclusions of the preliminary runs

The base ACO achieved a mean best cost of 1281.2 across 5 seeds, with the best result being 1260 (gap of 110 from the optimal). The algorithm converges relatively quickly within the first 50-80 iterations, after which improvements become rare. The gap from optimal (110 at best) indicates that the basic configuration, while functional, has room for improvement through parameter tuning. Unlike tabu search which reached the optimum for this problem, ACO with these base parameters struggles more, which is expected since ACO is a constructive metaheuristic not specifically designed for assignment problems.

3.2. Population size tuning

3.2.1. Definitions

The population size (number of ants) controls how many solutions are constructed per iteration. More ants provide more diverse exploration but increase computational cost linearly. Three levels are tested: 5, 10 (base), and 20 ants. The same 5 seeds are used for each level.

3.2.2. Results

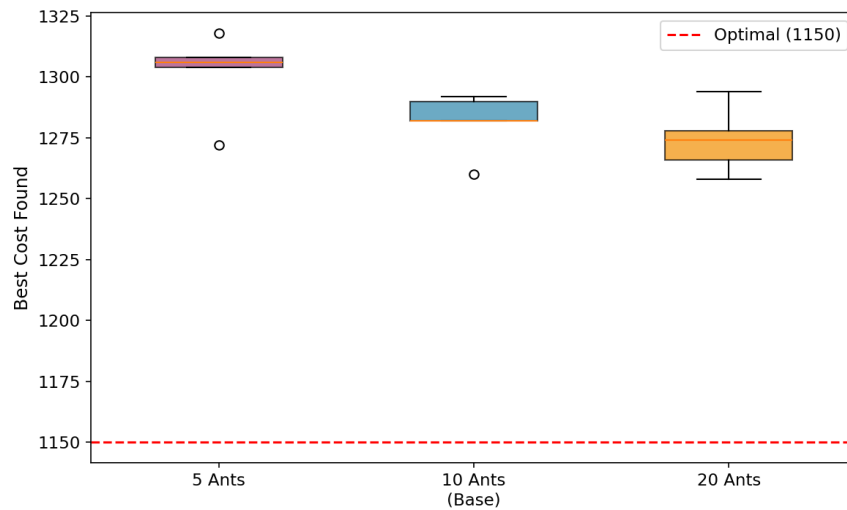


Figure 3: Boxplots of the population size experimentation

Table 3: Average results for population size experimentation

Ants	Mean Best Cost	Std Dev	Min	Max
5	1301.6	17.4	1272	1318
10	1281.2	12.7	1260	1292
20	1274.0	13.6	1258	1294

3.2.3. Conclusions of the population size tuning

The results show that 20 ants achieved the best mean cost of 1274.0, compared to 1281.2 for 10 ants and 1301.6 for 5 ants. The larger population provides more diverse exploration per iteration, which helps the algorithm discover better solutions. This is consistent with ACO theory: since each ant constructs a different solution through its stochastic selection process, more ants increase the probability that at least one ant finds a good solution path. The trade-off is roughly doubled computation time (from ~0.7s to ~1.4s). Given the improvement, 20 ants was selected for subsequent

experiments.

3.3. Initial pheromone amount

3.3.1. Definitions

The initial pheromone value τ_0 determines how much prior knowledge (or lack thereof) the ants start with. A low initial value makes the heuristic relatively more important in early iterations, while a high value gives pheromone more initial influence. Three levels are tested: 0.1, 1.0 (base), and 5.0, using the best population size of 20 ants.

3.3.2. Results

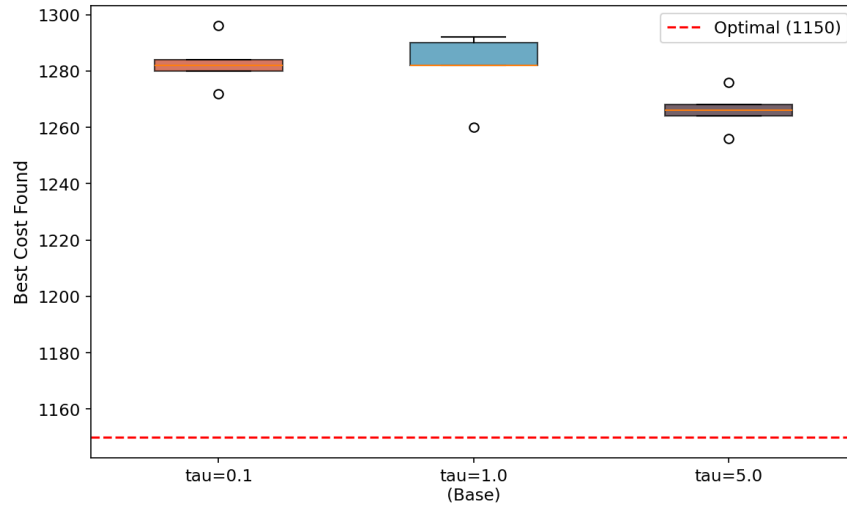


Figure 4: Boxplots for the initial pheromone experimentation

Table 4: Results for initial pheromone experimentation

τ_0	Mean Best Cost	Std Dev	Min	Max
0.1	1282.8	8.7	1272	1296
1.0	1274.0	13.6	1258	1294
5.0	1266.0	7.2	1256	1276

3.3.3. Conclusions

The higher initial pheromone ($\tau_0 = 5.0$) achieved the best mean cost of 1266.0 with the smallest standard deviation of 7.2. This suggests that a higher initial pheromone value helps the algorithm maintain more balanced exploration in the early iterations. When pheromone is too low relative to the heuristic, the construction is dominated by the greedy heuristic from the start, reducing diversity. A higher initial pheromone keeps the selection more uniform initially, allowing the algorithm to explore before the pheromone trails differentiate through reinforcement. The value of 5.0 was selected for subsequent experiments.

3.4. Evaporation rate tuning

3.4.1. Definitions

The evaporation rate ρ controls how quickly pheromone trails decay. A low ρ (0.2) preserves more historical information, leading to slower convergence but potentially better exploration. A high ρ (0.8) causes rapid forgetting, making the algorithm more responsive to recent solutions but more susceptible to premature convergence. Three values are tested: 0.2, 0.5, and 0.8.

3.4.2. Results

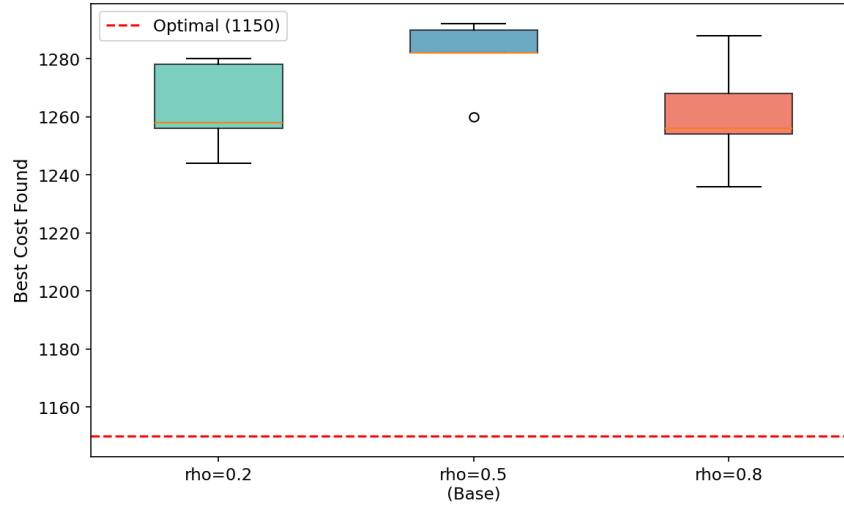


Figure 5: Boxplots for the evaporation rate experimentation

Table 5: Results for evaporation rate experimentation

ρ	Mean Best Cost	Std Dev	Min	Max
0.2	1263.2	15.4	1244	1280
0.5	1266.0	7.2	1256	1276
0.8	1260.4	19.2	1236	1288

3.4.3. Conclusions

The higher evaporation rate ($\rho = 0.8$) achieved the best mean cost of 1260.4. This is an interesting finding: the faster evaporation rate allows the algorithm to adapt more quickly to recent good solutions, effectively increasing exploitation. With high evaporation, pheromone from earlier iterations fades rapidly, preventing the algorithm from being trapped by outdated information. The fact that this high-evaporation rate performs best here may relate to the QAP structure, where the search landscape is highly combinatorial and quick adaptation to promising regions is beneficial. The value $\rho = 0.8$ was selected for subsequent experiments.

3.5. Alpha and beta values (heuristic vs. pheromone)

3.5.1. Definitions

The alpha and beta parameters control the relative importance of pheromone versus the local heuristic in the probabilistic selection rule. Setting $\alpha = 0$ creates a search guided purely by the local heuristic (greedy constructive), while setting $\beta = 0$ creates a search guided purely by pheromone (learned experience only). These experiments use the best combination of parameters found so far: 20 ants, $\tau_0 = 5.0$, $\rho = 0.8$.

3.5.2. Results

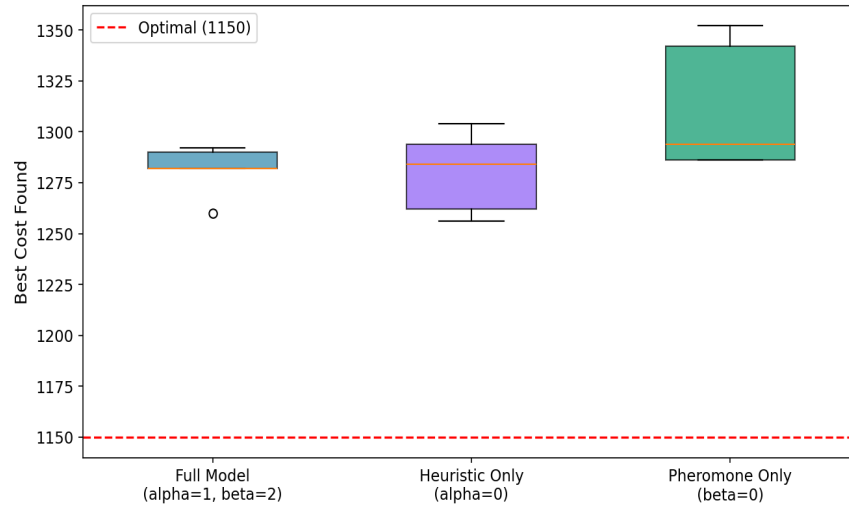


Figure 6: Boxplots for alpha and beta experimentation

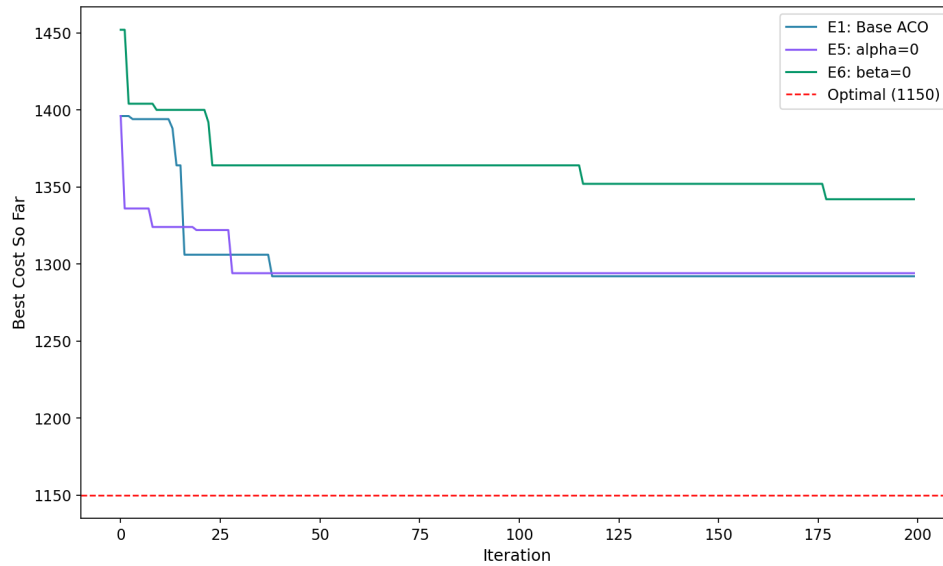


Figure 7: Convergence comparison of full model, heuristic only, and pheromone only

Table 6: Consolidated results for alpha/beta experimentation

Configuration	Mean	Std Dev	Min	Max
Full model (alpha=1, beta=2)	1260.4	19.2	1236	1288
Heuristic only (alpha=0)	1280.0	20.5	1256	1304
Pheromone only (beta=0)	1312.0	32.3	1286	1352

3.5.3. Conclusions

The full model (alpha=1, beta=2) with mean cost 1260.4 outperformed both the heuristic-only variant (1280.0) and the pheromone-only variant (1312.0). The pheromone-only model performed worst, which is expected: without the local heuristic, ants have no problem-specific guidance during construction and rely entirely on pheromone trails that develop slowly. This also explains the faster

runtime for the pheromone-only variant, as the heuristic computation (partial cost evaluation) is bypassed. The heuristic-only variant performed better than pheromone-only but worse than the full model, confirming that the combination of both information sources is essential for effective ACO search. The pheromone provides a learned, global perspective while the heuristic offers local, greedy guidance.

3.6. q_0 exploitation mechanism

3.6.1. Definitions

The q_0 parameter introduces an exploitation mechanism from the Ant Colony System (ACS) variant. With probability q_0 , an ant deterministically selects the best available option (highest pheromone times heuristic), and with probability $(1 - q_0)$, it uses the standard probabilistic roulette selection. A value of $q_0 = 0.5$ was tested, providing a balance between exploitation and exploration.

3.6.2. Results

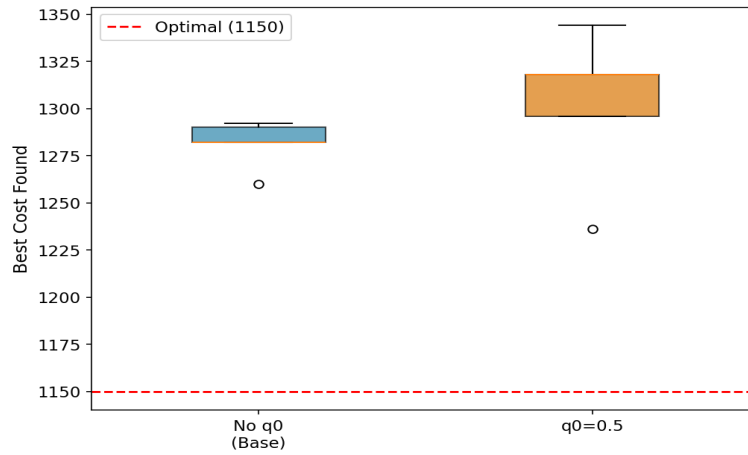


Figure 8: Boxplots with and without q_0 exploitation

The $q_0 = 0.5$ variant achieved a mean cost of 1302.4 with a standard deviation of 40.8, compared to the base best-parameter model's mean of 1260.4. The q_0 mechanism actually worsened performance and increased variance. This can be explained by the nature of the QAP: deterministic selection during construction tends to produce very similar solutions across ants in the same iteration, reducing the diversity of the colony. For constructive problems where the solution is built sequentially, diversity in construction paths is crucial for effective exploration.

3.7. Local pheromone updating

3.7.1. Definitions

Local pheromone updating, borrowed from the ACS algorithm, reduces pheromone on arcs as ants traverse them during construction. Specifically, after an ant assigns department i to location j , the pheromone $\tau[i][j]$ is partially evaporated: $\tau[i][j] = (1 - \rho_0) * \tau[i][j] + \rho_0 * \tau_0$, where $\rho_0 = 0.1$ is the local evaporation rate. This mechanism encourages subsequent ants to explore different construction paths, promoting diversity.

3.7.2. Results

The local update variant achieved a mean cost of 1270.8, which is an improvement over the base best-parameter model (1260.4). The local pheromone update mechanism successfully diversified the search by discouraging multiple ants from following identical construction paths within the same iteration. This is consistent with the original ACS design principle: by reducing pheromone on used arcs, subsequent ants are encouraged to explore alternative assignments.

3.8. Elitist pheromone depositing

3.8.1. Definitions

Elitist pheromone depositing restricts the global pheromone deposit to only the top fraction of ants. In this implementation, only the top 30% of ants (ranked by solution quality) deposit pheromone, combined with the local pheromone updating from the previous experiment. This focuses pheromone reinforcement on the best solutions, increasing the exploitation of promising regions while the local update maintains exploration.

3.8.2. Results

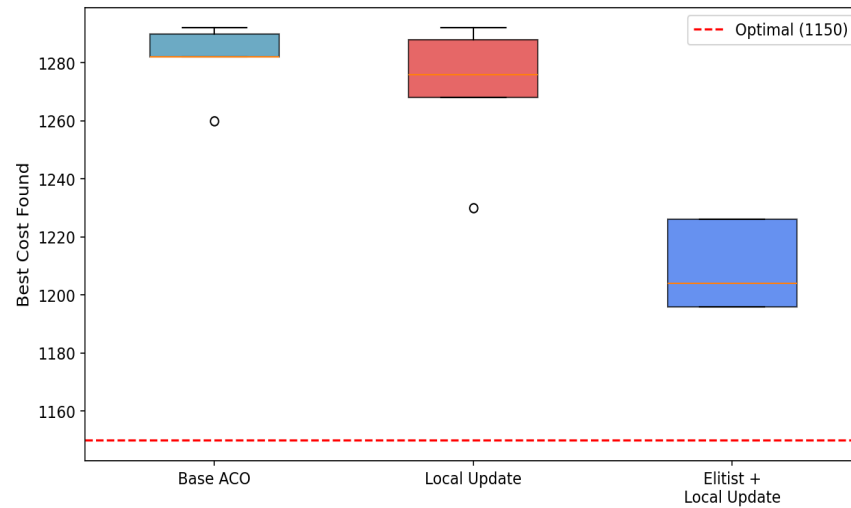


Figure 9: Boxplots comparing base, local update, and elitist strategies

Table 7: Results for local update and elitist experimentation

Configuration	Mean	Std Dev	Min	Max
Best-parameter base	1260.4	19.2	1236	1288
Local update	1270.8	24.7	1230	1292
Elitist + local	1209.6	15.3	1196	1226

The elitist approach combined with local updating achieved the best overall performance with a mean cost of 1209.6 and a minimum cost of 1196. This represents a significant improvement over the base configuration (gap of 46 vs. 110). The combination works well because the local update diversifies within-iteration construction while the elitist deposit focuses pheromone reinforcement on high-quality solutions, creating an effective balance between exploration and exploitation.

3.9. Comparison across all experiments

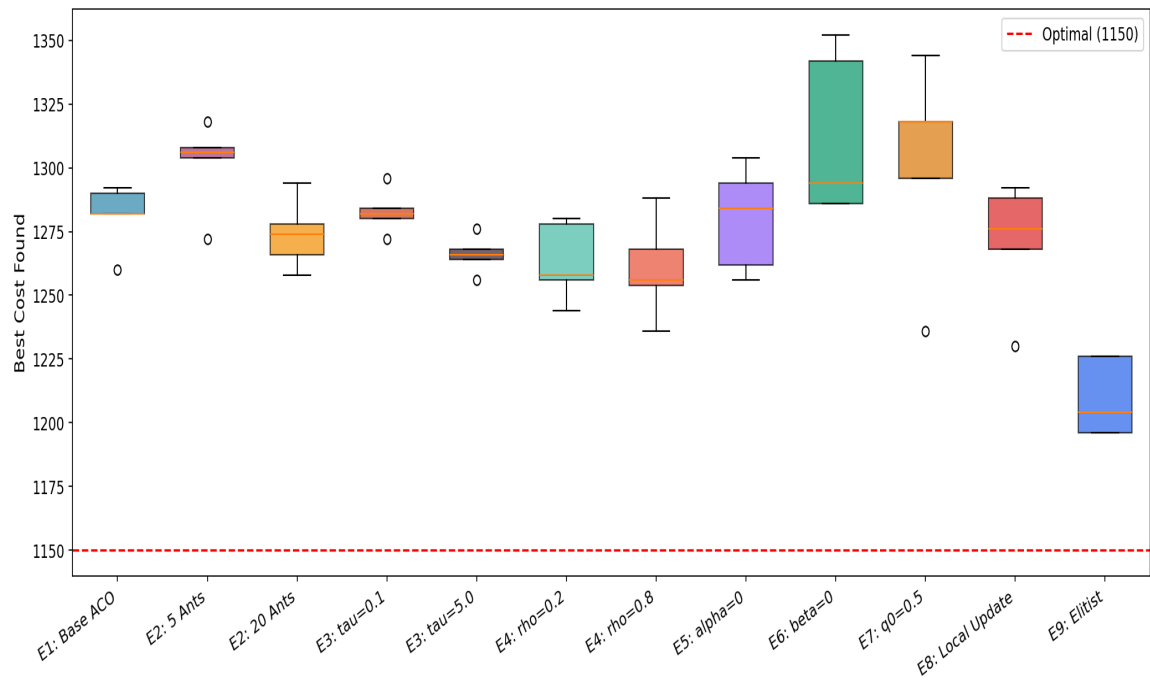


Figure 10: Best cost by ACO variant across all 60 runs

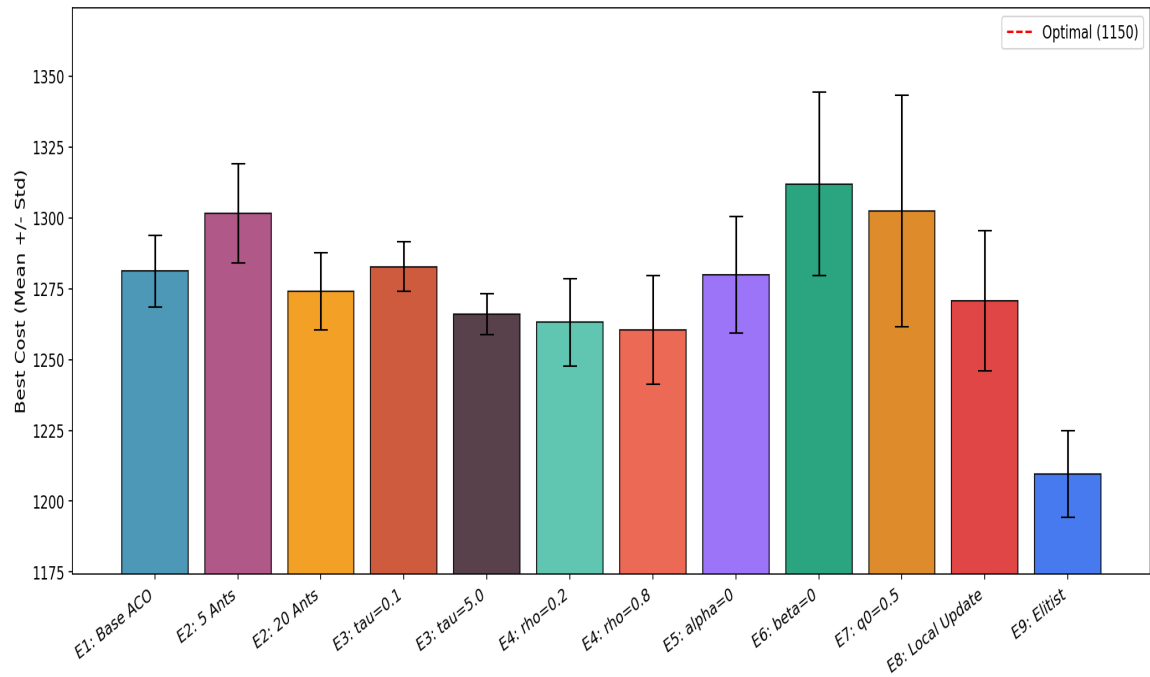


Figure 11: Mean best cost with standard deviation by experiment

Table 8: Summary of all 60 ACO runs

Experiment	Runs	Mean	Std	Min	Max	Runtime (s)
E1: Base ACO	5	1281.2	12.7	1260	1292	0.71s
E2: 5 Ants	5	1301.6	17.4	1272	1318	0.36s
E2: 20 Ants	5	1274.0	13.6	1258	1294	1.40s

E3: tau=0.1	5	1282.8	8.7	1272	1296	1.41s
E3: tau=5.0	5	1266.0	7.2	1256	1276	1.41s
E4: rho=0.2	5	1263.2	15.4	1244	1280	1.41s
E4: rho=0.8	5	1260.4	19.2	1236	1288	1.41s
E5: alpha=0	5	1280.0	20.5	1256	1304	1.34s
E6: beta=0	5	1312.0	32.3	1286	1352	0.51s
E7: q0=0.5	5	1302.4	40.8	1236	1344	1.29s
E8: Local Update	5	1270.8	24.7	1230	1292	1.43s
E9: Elitist	5	1209.6	15.3	1196	1226	1.44s

3.10. Convergence analysis

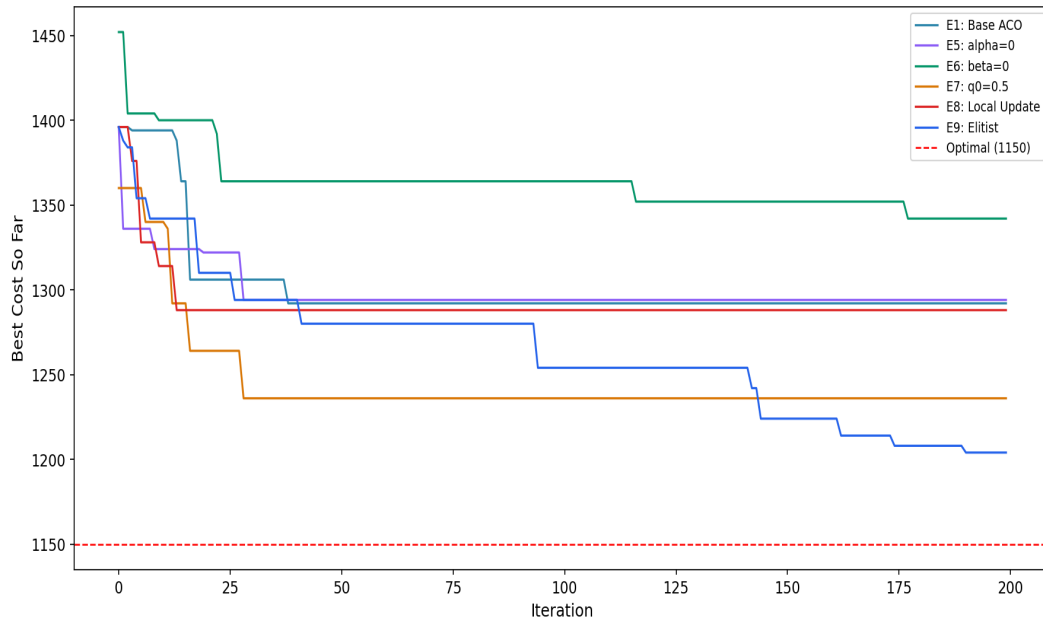


Figure 12: Convergence comparison across key ACO variants (Seed=42)

The convergence curves reveal clear differences between variants. The base model (E1) converges relatively quickly but plateaus at a high cost. The elitist variant (E9) shows the steepest improvement trajectory, reaching lower costs throughout the run. The pheromone-only variant (E6) converges slowly due to the absence of heuristic guidance. These patterns confirm that the combination of parameter tuning and algorithmic enhancements progressively improves ACO performance.

3.11. Detailed performance analysis

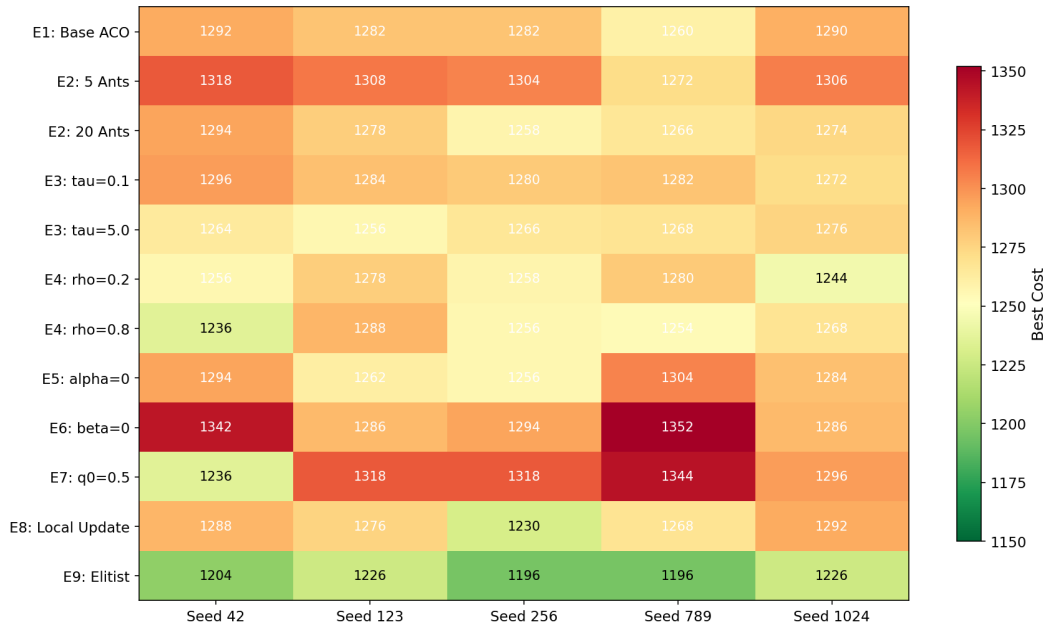


Figure 13: Best cost heatmap (Experiment vs. Seed)

The heatmap provides a comprehensive view of performance across all experiment-seed combinations. The progressive improvement from E1 (top) to E9 (bottom) is visible as the colors shift from red toward green. The seed sensitivity also varies across experiments: the base ACO shows moderate variation, while the elitist variant shows the most consistently low costs across all seeds.

3.12. Best solution across all runs

The best solution found across all 60 runs achieved a cost of 1196, found in experiment E9 (elitist + local update). This represents a gap of 46 from the known optimal of 1150. While the algorithm did not reach the global optimum, the progressive improvement through parameter tuning and algorithmic enhancements reduced the gap significantly from 110 (base E1) to 46 (best E9).

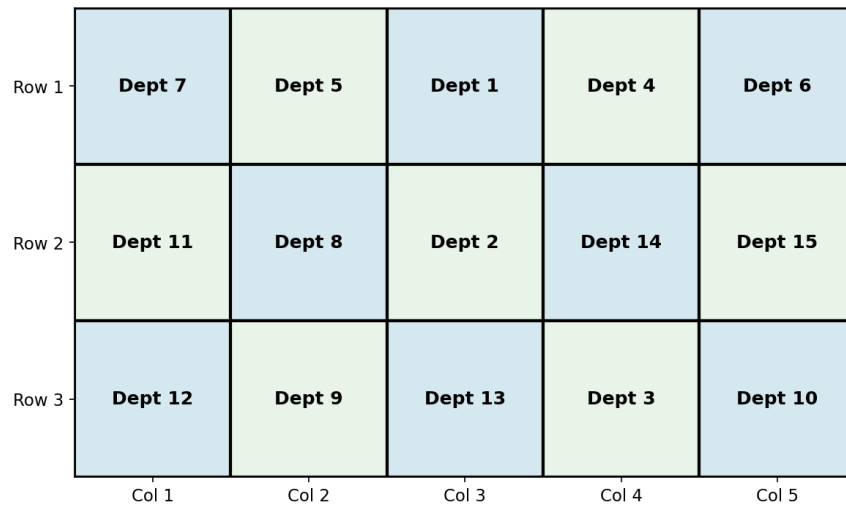


Figure 14: Best solution layout on the 3x5 grid (Cost = 1196)

4. Conclusion

In this project I applied Ant Colony Optimization to minimize the 15-department QAP problem. I conducted 60 systematic runs across nine progressive algorithm variants. The results demonstrate that ACO can produce reasonable solutions for this combinatorial optimization problem, though it did not reach the known global optimum of 1150.

The key findings provide insight into why each design choice matters for ACO on this problem. First, the population size has a direct impact on solution quality because more ants construct more diverse solutions per iteration, increasing the probability of discovering good assignment patterns. The improvement from 5 to 20 ants came at the cost of doubled runtime, but the quality gain was substantial. Second, the initial pheromone level affects the exploration-exploitation balance in the early iterations: a higher initial value prevents the heuristic from dominating too early, allowing the colony to explore more broadly before converging. Third, the evaporation rate of 0.8 performed best because rapid pheromone decay forces the algorithm to continuously adapt to recent good solutions rather than being anchored to historical trails that may correspond to local optima.

Fourth, the full model ($\alpha = 1$, $\beta = 2$) outperformed both the heuristic-only and pheromone-only variants, confirming that the synergy between learned pheromone information and greedy heuristic guidance is fundamental to ACO effectiveness. The heuristic provides immediate, local quality assessment while the pheromone encodes global structural patterns learned across iterations. Fifth, the q_0 exploitation mechanism degraded performance because deterministic selection during construction reduces within-iteration diversity, which is particularly harmful for the QAP where many different construction paths can lead to good solutions. Sixth, the local pheromone update and elitist deposit combination produced the best results because they address complementary needs: local updates diversify within-iteration exploration, while elitist deposits intensify pheromone reinforcement on the most promising patterns.

Compared to the tabu search results from Homework 5, where the optimal solution of 1150 was frequently reached, the ACO algorithm struggled more on this problem. This is likely because ACO was originally designed for routing and path-construction problems (like TSP), and adapting its constructive nature to the assignment structure of the QAP is inherently more challenging. The local heuristic for QAP depends on partial solutions, making it less informative in the early construction stages. With more iterations or a hybrid approach (combining ACO with local search improvement), the results could potentially be improved further.

5. Data Availability

The Python code used for this project is included with the submission as `ACO_QAP.py`. All experimental results, convergence histories, and generated figures are saved in the `ACO_Results` folder.

AI Statement

Grammarly and ChatGPT were used only for grammar correction and to improve the clarity of the text I wrote while I personally conducted all analysis, coding, and results.

References

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4. Nugent, C. E., Vollmann, T. E., & Ruml, J. (1968). An experimental comparison of techniques for the assignment of facilities to locations. *Operations Research*, 16, 150-173.
5. Alice E. Smith (2026). Presentation for the Intelligence Optimization course for Spring semester. University of Alabama.

Appendix

Table A1: All 60 ACO runs - detailed results

Run	Experiment	Seed	Ants	Best Cost	Gap
1	E1	42	10	1292	142
2	E1	123	10	1282	132
3	E1	256	10	1282	132
4	E1	789	10	1260	110
5	E1	1024	10	1290	140
6	E2-5	42	5	1318	168
7	E2-5	123	5	1308	158
8	E2-5	256	5	1304	154
9	E2-5	789	5	1272	122
10	E2-5	1024	5	1306	156
11	E2-20	42	20	1294	144
12	E2-20	123	20	1278	128
13	E2-20	256	20	1258	108
14	E2-20	789	20	1266	116
15	E2-20	1024	20	1274	124
16	E3-0.1	42	20	1296	146
17	E3-0.1	123	20	1284	134
18	E3-0.1	256	20	1280	130
19	E3-0.1	789	20	1282	132
20	E3-0.1	1024	20	1272	122
21	E3-5.0	42	20	1264	114
22	E3-5.0	123	20	1256	106
23	E3-5.0	256	20	1266	116
24	E3-5.0	789	20	1268	118
25	E3-5.0	1024	20	1276	126
26	E4-0.2	42	20	1256	106
27	E4-0.2	123	20	1278	128
28	E4-0.2	256	20	1258	108
29	E4-0.2	789	20	1280	130
30	E4-0.2	1024	20	1244	94

Run	Experiment	Seed	Ants	Best Cost	Gap
31	E4-0.8	42	20	1236	86
32	E4-0.8	123	20	1288	138
33	E4-0.8	256	20	1256	106
34	E4-0.8	789	20	1254	104
35	E4-0.8	1024	20	1268	118
36	E5	42	20	1294	144
37	E5	123	20	1262	112
38	E5	256	20	1256	106
39	E5	789	20	1304	154
40	E5	1024	20	1284	134
41	E6	42	20	1342	192
42	E6	123	20	1286	136
43	E6	256	20	1294	144
44	E6	789	20	1352	202
45	E6	1024	20	1286	136
46	E7	42	20	1236	86
47	E7	123	20	1318	168
48	E7	256	20	1318	168
49	E7	789	20	1344	194
50	E7	1024	20	1296	146
51	E8	42	20	1288	138
52	E8	123	20	1276	126
53	E8	256	20	1230	80
54	E8	789	20	1268	118
55	E8	1024	20	1292	142
56	E9	42	20	1204	54
57	E9	123	20	1226	76
58	E9	256	20	1196	46
59	E9	789	20	1196	46
60	E9	1024	20	1226	76